Requirements Coverage Report - COMP3211 PIM

Group: 89; member names and student IDs await human input. **Revision:** Usability release, 2026-10-23. The candidate manifest identifies the current files. **Scope:** US1–US11, FR-01–FR-54 and NFR-01–NFR-05. Both coverage PDFs are directly in the packaged `03_Implementation` source root.

All 86 test methods pass on Python 3.11.9, 3.12.3 and 3.13.5. Subtests are not counted separately. Fresh full logs are `usability-verification-2026-10-23-python311.txt`, `python312.txt` and `python313.txt` (same prefix), under repository `docs/reports/` and packaged `evidence/`. Implemented means source and named tests support the stated cases; it is not proof for all inputs. NFR-01 retains the exact-environment evidence gap below.

Named test classes are in the model, controller and storage test directories, plus real-process acceptance tests. `UsabilityTests` covers unsaved protection and help; `SearchPositionTests` covers source positions. Terminal/process/evidence-tool tests also retain the peer-comparison hardening. Earlier dated reports describe historical revisions.

1. Appendix B user stories

Story	Status	Current source file and symbol	Test or manual evidence	Notes
US1	Implemented	`model/manager.py::PIMManager._create`, `.list_all`; `controller/cli.py::CommandController._dispatch`	`ManagerTests.test_create_all_types_and_monotonic_ids`; `AcceptanceTests.test_all_stories_across_save_and_restart`	One collection holds all four types.
US2	Implemented	`model/records.py::Note`; `model/manager.py::PIMManager.create_note`	`RecordTests.test_four_types_trim_values_and_keep_mobile_text`; `AcceptanceTests.test_all_stories_across_save_and_restart`	Note text.
US3	Implemented	`model/records.py::Task`; `model/manager.py::PIMManager.create_task`	`ManagerTests.test_create_all_types_and_monotonic_ids`	Description and deadline.
US4	Implemented	`model/records.py::Event`; `model/manager.py::PIMManager.create_event`	`ManagerTests.test_create_all_types_and_monotonic_ids`	Description, start, alarm.
US5	Implemented	`model/records.py::Contact`; `model/manager.py::PIMManager.create_contact`	`RecordTests.test_four_types_trim_values_and_keep_mobile_text`	Name, address, mobile number.
US6	Implemented	`model/manager.py::PIMManager.update`; `controller/cli.py::CommandController._dispatch`	`ManagerTests.test_get_list_update_delete_and_invalid_ids`; `SearchIntegrationTests.test_search_orders_records_and_does_not_mutate`	One-field update.
US7	Implemented	`model/search.py::parse_criterion`, `Criterion.matches`; `model/manager.py::PIMManager.search`	`SearchTests.test_type_and_all_text_fields_are_case_insensitive`, `.test_each_time_field_and_comparison_operator`, `.test_boolean_precedence_parentheses_negation_and_absent_field`	All official atom and connector categories.

Story	Status	Current source file and symbol	Test or manual evidence	Notes
US8	Implemented	`model/manager.py::PIMManager.get`, `.list_all`; `controller/cli.py::_details`, `CommandController._dispatch`	`CLITests.test_add_list_show_update_delete_and_help`	One/all detail output.
US9	Implemented	`model/manager.py::PIMManager.delete`; `controller/cli.py::CommandController._dispatch`	`ManagerTests.test_get_list_update_delete_and_invalid_ids`	Specified ID only.
US10	Implemented	`storage/pim_file.py::save_pim`; `controller/cli.py::CommandController._dispatch`	`PersistenceTests.test_round_trip_four_types_and_preserve_higher_counter`	`.pim` save.
US11	Implemented	`storage/pim_file.py::load_pim`; `model/manager.py::PIMManager.from_rows`	`PersistenceTests.test_round_trip_four_types_and_preserve_higher_counter`	`.pim` load after restart.

2. SRS functional requirements

Source paths in this table are relative to `src/`. A symbol after `::` is the exact current class, function, method, or module constant. All referenced checks were rerun on the current usability revision. These checks cover the stated cases; they are not proof over all possible inputs.

ID	Status	Current source file and symbol	Test or manual evidence	Notes
FR-01	Implemented	`model/manager.py::PIMManager._create`, `.list_all`	`ManagerTests.test_create_all_types_and_monotonic_ids`	Four types coexist.
FR-02	Implemented	`model/records.py::Note`; `model/manager.py::PIMManager.create_note`	`RecordTests.test_four_types_trim_values_and_keep_mobile_text`	Required text.
FR-03	Implemented	`model/records.py::Task`; `model/manager.py::PIMManager.create_task`	`ManagerTests.test_create_all_types_and_monotonic_ids`	Required description and deadline.
FR-04	Implemented	`model/records.py::Event`; `model/manager.py::PIMManager.create_event`	`ManagerTests.test_create_all_types_and_monotonic_ids`; `ModelRequirementTests.test_past_times_and_alarm_after_start_are_accepted`	Both historical event times and alarm-after-start values are asserted.
FR-05	Implemented	`model/records.py::Contact`; `model/manager.py::PIMManager.create_contact`	`RecordTests.test_four_types_trim_values_and_keep_mobile_text`	Required fields and `+00123` preserved.
FR-06	Implemented	`model/validation.py::validate_text`; `model/records.py::Note.__post_init__`, `Task.__post_init__`, `Event.__post_init__`, `Contact.__post_init__`	`RecordTests.test_four_types_trim_values_and_keep_mobile_text`, `.test_invalid_direct_construction`, `.test_unicode_line_separators_are_not_single_line_text`; `ManagerTests.test_invalid_creation_does_not_consume_id`	Text fields trim; time fields must match FR-07 exactly. Tests sample invalid classes across types.

ID	Status	Current source file and symbol	Test or manual evidence	Notes
FR-07	Implemented	`model/validation.py::parse_local_time`, `validate_datetime`; `model/manager.py::PIMManager._create`	`RecordTests.test_external_time_has_exact_minute_format`, `.test_invalid_direct_construction`; `ModelRequirementTests.test_past_times_and_alarm_after_start_are_accepted`	Past-time acceptance now has a direct assertion; existing format/date rejection tests remain.
FR-08	Implemented	`model/records.py::Contact.__post_init__`; `model/validation.py::validate_text`	`RecordTests.test_four_types_trim_values_and_keep_mobile_text`	Mobile value is text; no regional pattern is applied.
FR-09	Implemented	`model/manager.py::PIMManager._create`, `.delete`, `.from_rows`	`ManagerTests.test_create_all_types_and_monotonic_ids`, `.test_from_rows_checks_schema_order_and_counters`	New IDs respect the session and loaded counters; a fresh process has no separate external ID history.
FR-10	Implemented	`model/manager.py::PIMManager._create`	`ManagerTests.test_invalid_creation_does_not_consume_id`	Counter advances after validated construction.
FR-11	Implemented	`model/manager.py::PIMManager._create`	`ManagerTests.test_create_all_types_and_monotonic_ids`	Repeated x data across types receives distinct IDs.
FR-12	Implemented	`controller/cli.py::_arguments`, `_count`, `CommandController._dispatch`	`CLITests.test_add_list_show_update_delete_and_help`, `.test_quote_escapes_argument_counts_ids_and_eof`; `AcceptanceTests.test_all_stories_across_save_and_restart`	Commands and counts dispatched.
FR-13	Implemented	`controller/cli.py::CommandController._dispatch`; `model/manager.py::_type_name`, `PIMManager.update`	`CLITests.test_add_list_show_update_delete_and_help`; `ManagerTests.test_get_list_update_delete_and_invalid_ids`; `CLIRequirementTests.test_all_command_forms_accept_mixed_case`	All command forms and four creation types run in mixed case; stored values retain case. The all-field update matrix also uses uppercase field names.
FR-14	Implemented	`controller/cli.py::_FIELDS`, `_count`, `CommandController._dispatch`	`CLITests.test_add_list_show_update_delete_and_help`, `.test_quote_escapes_argument_counts_ids_and_eof`; `ManagerTests.test_create_all_types_and_monotonic_ids`	Schema order and type-specific count.
FR-15	Implemented	`controller/cli.py::_arguments`	`CLITests.test_quote_escapes_argument_counts_ids_and_eof`, `.test_errors_do_not_mutate_and_next_command_runs`; `CLIRequirementTests.test_bad_quoting_and_escapes_recover_without_mutation`	Unmatched quotes and unsupported escapes reject the command; state and the next command are checked.
FR-16	Implemented	`model/manager.py::PIMManager.update`; `controller/cli.py::CommandController._dispatch`	`ManagerTests.test_get_list_update_delete_and_invalid_ids`; `SearchIntegrationTests.test_search_orders_records_and_does_not_mutate`; `ModelRequirementTests.test_every_mutable_field_changes_only_that_field`	Each of the nine mutable fields is updated; ID, type, other fields, other records and next ID are checked.

ID	Status	Current source file and symbol	Test or manual evidence	Notes
FR-17	Implemented	`model/manager.py::PIMManager.get`, `.update`	`ManagerTests.test_failed_update_is_atomic_and_forbidden_fields`	ID/type/absent field rejected without mutation.
FR-18	Implemented	`model/manager.py::PIMManager.list_all`; `controller/cli.py::_summary`, `CommandController._dispatch`	`CLITests.test_add_list_show_update_delete_and_help`	Ordered, filtered summaries.
FR-19	Implemented	`controller/cli.py::_details`, `CommandController._dispatch`	`CLITests.test_add_list_show_update_delete_and_help`; `CLIRequirementTests.test_show_each_type_prints_all_details`	Exact show-ID output is checked for each of the four types and all fields.
FR-20	Implemented	`model/manager.py::PIMManager.list_all`; `controller/cli.py::_details`, `CommandController._dispatch`	`CLITests.test_add_list_show_update_delete_and_help`	All-field show-ID assertions and model ordering tests support this path; show-all formatting/order also inspected in source. No exact whole show-all transcript assertion.
FR-21	Implemented	`controller/cli.py::CommandController._print_records`	`CLITests.test_search_compound_and_no_matches`; `CLIRequirementTests.test_empty_list_filter_and_show_all_have_clear_output`	Empty list, filtered list and show-all output is asserted, including an unmatched filter in a nonempty collection.
FR-22	Implemented	`model/manager.py::PIMManager.delete`	`ManagerTests.test_get_list_update_delete_and_invalid_ids`	Selected ID removed.
FR-23	Implemented	`controller/cli.py::_id`, `CommandController.run`; `model/manager.py::PIMManager.get`	`CLITests.test_quote_escapes_argument_counts_ids_and_eof`; `ManagerTests.test_get_list_update_delete_and_invalid_ids`; `AcceptanceTests.test_very_long_id_is_reported_and_session_continues`	Invalid/absent IDs recover.
FR-24	Implemented	`controller/cli.py::_HELP`, `CommandController._dispatch`	`CLITests.test_add_list_show_update_delete_and_help`; clean-checkout `help` smoke run	Help lists forms and grammar.
FR-25	Implemented	`controller/cli.py::CommandController._dispatch`, `.run`	`CLITests.test_quote_escapes_argument_counts_ids_and_eof`; `AcceptanceTests.test_all_stories_across_save_and_restart`	Exit and EOF.
FR-26	Implemented	`controller/cli.py::CommandController._dispatch`; `model/search.py::parse_criterion`	`SearchTests.test_invalid_expression_is_rejected_whole`; `CLITests.test_search_compound_and_no_matches`	Empty and valid search paths.
FR-27	Implemented	`model/manager.py::PIMManager.search`; `model/search.py::Criterion.matches`	`SearchIntegrationTests.test_search_orders_records_and_does_not_mutate`	Searches current collection.
FR-28	Implemented	`model/search.py::TypeCriterion`, `_Parser._atom`	`SearchTests.test_type_and_all_text_fields_are_case_insensitive`, `.test_invalid_expression_is_rejected_whole`	Canonical types.

ID	Status	Current source file and symbol	Test or manual evidence	Notes
FR-29	Implemented	`model/search.py::TextCriterion`, `_Parser._atom`	`SearchTests.test_type_and_all_text_fields_are_case_insensitive`, `.test_invalid_expression_is_rejected_whole`	Five fields; empty/unquoted rejected.
FR-30	Implemented	`model/search.py::TextCriterion.matches`	`SearchTests.test_type_and_all_text_fields_are_case_insensitive`	Case-insensitive substring.
FR-31	Implemented	`model/search.py::TimeCriterion`, `_Parser._atom`	`SearchTests.test_each_time_field_and_comparison_operator`, `.test_invalid_expression_is_rejected_whole`	Nine field/operator pairs.
FR-32	Implemented	`model/search.py::TimeCriterion.matches`	`SearchTests.test_each_time_field_and_comparison_operator`; `ModelRequirementTests.test_time_comparisons_cross_calendar_boundaries`	All three time fields and operators are checked at day, month, leap-day and year boundaries, including equality and false outcomes.
FR-33	Implemented	`model/search.py::AndCriterion`, `_evaluate`	`SearchTests.test_boolean_precedence_parentheses_negation_and_absent_field`; `SearchIntegrationTests.test_search_orders_records_and_does_not_mutate`; `ModelRequirementTests.test_and_or_full_truth_tables`	All four Boolean input combinations are asserted for direct AND nodes and parsed AND expressions.
FR-34	Implemented	`model/search.py::OrCriterion`, `_evaluate`	`SearchTests.test_boolean_precedence_parentheses_negation_and_absent_field`; `SearchIntegrationTests.test_search_orders_records_and_does_not_mutate`; `ModelRequirementTests.test_and_or_full_truth_tables`	All four Boolean input combinations are asserted for direct OR nodes and parsed OR expressions.
FR-35	Implemented	`model/search.py::NotCriterion`, `_evaluate`	`SearchTests.test_boolean_precedence_parentheses_negation_and_absent_field`	Negation and double negation.
FR-36	Implemented	`model/search.py::_Parser.parse`, `_evaluate`	`SearchTests.test_boolean_precedence_parentheses_negation_and_absent_field`, `.test_deep_valid_expressions_evaluate_without_recursion`; `BooleanCompositionTests.test_three_atom_compositions_against_independent_truth_tables`	Grouping and precedence.
FR-37	Implemented	`model/search.py::TextCriterion.matches`, `TimeCriterion.matches`, `_evaluate`	`SearchTests.test_boolean_precedence_parentheses_negation_and_absent_field`	Missing field yields false; negation tested.
FR-38	Implemented	`model/search.py::_tokenize`, `_Parser.parse`, `_Parser._atom`, `parse_criterion`; `controller/cli.py::CommandController.run`	`SearchTests.test_invalid_expression_is_rejected_whole`; `CLITests.test_errors_do_not_mutate_and_next_command_runs`	Whole expression rejected before result display.
FR-39	Implemented	`model/manager.py::PIMManager.search`; `controller/cli.py::_summary`, `CommandController._print_records`	`SearchIntegrationTests.test_search_orders_records_and_does_not_mutate`; `CLITests.test_search_compound_and_no_matches`	Ordering asserted at model boundary; CLI test asserts summary content. Formatting additionally inspected in source.

ID	Status	Current source file and symbol	Test or manual evidence	Notes
FR-40	Implemented	`controller/cli.py::CommandController._print_records`	`CLITests.test_search_compound_and_no_matches`	`No matching PIRs.`
FR-41	Implemented	`storage/pim_file.py::_path`, `save_pim`; `controller/cli.py::CommandController._dispatch`	`PersistenceTests.test_round_trip_four_types_and_preserve_higher_counter`, `.test_wrong_extension_and_failed_save_preserve_existing_bytes`	Exact lowercase `.pim`.
FR-42	Implemented	`storage/pim_file.py::_row`, `save_pim`	`PersistenceTests.test_round_trip_four_types_and_preserve_higher_counter`	UTF-8 JSON, version 1, exact fields.
FR-43	Implemented	`storage/pim_file.py::save_pim`	`PersistenceTests.test_wrong_extension_and_failed_save_preserve_existing_bytes`	Replacement failure is injected and old bytes checked; serialization/write error paths inspected in source, not separately fault-injected.
FR-44	Implemented	`storage/pim_file.py::_path`, `load_pim`; `controller/cli.py::CommandController._dispatch`	`PersistenceTests.test_round_trip_four_types_and_preserve_higher_counter`, `.test_reject_missing_invalid_and_duplicate_json`; `CLITests.test_save_load_swap_and_failure_preserves_active_manager`	Validation before active-manager swap; `MalformedSnapshotTests.test_excessive_json_nesting_reports_format_error` verifies excessive nesting produces FORMAT.
FR-45	Implemented	`storage/pim_file.py::_unique_object`, `_reject_constant`, `load_pim`; `model/manager.py::PIMManager.from_rows`	`PersistenceTests.test_reject_missing_invalid_and_duplicate_json`; `ManagerTests.test_from_rows_checks_schema_order_and_counters`; `SnapshotRequirementTests.test_missing_extra_keys_and_invalid_fields_are_rejected_atomically`	Each root/record key is removed, extra keys added and every data field given invalid values; malformed loads preserve live records and next ID. Representative invalid classes are tested, not all possible files.
FR-46	Implemented	`model/manager.py::PIMManager.from_rows`; `controller/cli.py::CommandController._dispatch`	`PersistenceTests.test_round_trip_four_types_and_preserve_higher_counter`; `CLITests.test_save_load_swap_and_failure_preserves_active_manager`	Replace collection, keep higher counter.
FR-47	Implemented	`storage/pim_file.py::load_pim`; `controller/cli.py::CommandController._dispatch`	`CLITests.test_save_load_swap_and_failure_preserves_active_manager`; `AcceptanceTests.test_invalid_input_recovers_in_real_process`	Failed load leaves active manager; `MalformedSnapshotTests.test_excessive_nesting_preserves_state_and_session` checks deep JSON, subsequent commands and next ID.
FR-48	Implemented	`controller/cli.py::CommandController._path`	`CLITests.test_save_load_swap_and_failure_preserves_active_manager`; `AcceptanceTests.test_all_stories_across_save_and_restart`	Tests set a process working directory and use relative paths.
FR-49	Implemented	`controller/cli.py::CommandController.run`; `storage/pim_file.py::PersistenceError`	`CLITests.test_errors_do_not_mutate_and_next_command_runs`, `.test_tab_separated_search_and_invalid_path_recover`; `AcceptanceTests.test_invalid_input_recovers_in_real_process`, `.test_very_long_id_is_reported_and_session_continues`	Categories and prompt recovery; `PromptVisibilityTests.test_prompt_is_visible_before_waiting_for_each_command` verifies buffered prompt visibility.

ID	Status	Current source file and symbol	Test or manual evidence	Notes
FR-50	Implemented	`model/manager.py::PIMManager._create`, `.update`, `.delete`; `storage/pim_file.py::save_pim`, `load_pim`; `controller/cli.py::CommandController._dispatch`	`ManagerTests.test_invalid_creation_does_not_consume_id`, `.test_failed_update_is_atomic_and_forbidden_fields`; `CLITests.test_errors_do_not_mutate_and_next_command_runs`, `.test_save_load_swap_and_failure_preserves_active_manager`; `CLIRequirementTests.test_each_failed_operation_preserves_state_and_returns_to_prompt`	Failed create/update/delete/search/save/load and every documented error category preserve records and next ID. Failed replacement preserves destination bytes; the next valid command succeeds.
FR-51	Implemented	`controller/cli.py::CommandController._dispatch`, `._warn_unsaved`	`tests/controller/test_usability.py`	Successful/failed operations, read-only commands and injected existing records preserve the intended dirty state.
FR-52	Implemented	`controller/cli.py::CommandController._dispatch`	`tests/controller/test_usability.py`; `tests/test_acceptance.py`	Protected/forced loads, force=false, malformed options, failure state and counter retention.
FR-53	Implemented	`controller/help.py::COMMAND_HELP`, `command_help`; controller dispatch	`tests/controller/test_usability.py`	All ten topics, case, usage/examples and recoverable errors.
FR-54	Implemented	`model/search.py::_tokenize`, `_Parser`; `model/errors.py::SearchSyntaxError`; `controller/cli.py::_arguments`	`tests/model/test_search_positions.py`; `tests/controller/test_usability.py`; `tests/test_acceptance.py`	Lexical/syntax/semantic/Unicode/EOF offsets and next-command recovery.

FR-24 now includes focused-help evidence in FR-53; FR-25 includes dirty exit/EOF warning checks in FR-51. FR-44 and FR-46 loads are subject to FR-52; invalid-file tests explicitly force loading when dirty so they still reach storage validation. FR-49 positioned diagnostics are checked by FR-54.

3. SRS non-functional requirements

ID	Status	Current source file and symbol	Test or manual evidence	Notes
NFR-01	Implemented; evidence gap	`main.py::main`; `controller/cli.py::CommandController.run`	`AcceptanceTests.test_all_stories_across_save_and_restart`	Local CLI runs; a network-disabled/no-graphical-session environment was not separately recorded.
NFR-02	Verified on listed versions	`main.py::main`; all application packages	86 tests including real-process restart acceptance on Python 3.11.9, 3.12.3 and 3.13.5; the three usability-verification logs listed above	Windows 11. The isolated 3.11.9 runtime contains no third-party packages. This verifies the listed versions, not every future Python release.
NFR-03	Implemented	`main.py::main`; `model/manager.py::PIMManager`; `controller/cli.py::CommandController`; `storage/pim_file.py::save_pim`	Fresh AST audit of 49 source imports; `usability-import-audit-2026-10-23.txt`; Python 3.11 launch and suite with site packages disabled	Whole-source import audit found only standard-library or local imports.

ID	Status	Current source file and symbol	Test or manual evidence	Notes
NFR-04	Implemented	`model/records.py::Note`, `Task`, `Event`, `Contact`; `model/manager.py::PIMManager`; `model/search.py::parse_criterion`	`ManagerTests.test_create_all_types_and_monotonic_ids`; `SearchIntegrationTests.test_search_orders_records_and_does_not_mutate`; fresh import audit	Model behavior resides in separate `model` package.
NFR-05	Implemented	`controller/cli.py::CommandController`; `storage/pim_file.py::save_pim`, `load_pim`	`CLITests.test_add_list_show_update_delete_and_help`; `PersistenceTests.test_round_trip_four_types_and_preserve_higher_counter`; `docs/design/Design_Document.md`	Distinct command and file components.

4. Evidence and remaining gaps

The usability revision passes **86 test methods** on all three listed Python versions. The controller and model changes add unsaved protection, help topics and positioned diagnostics. Existing schemas and persistence rules remain unchanged; the candidate manifest identifies the synchronized payload.

NFR-01's exact network-disabled/no-graphical-session environment remains unverified: execution used terminal streams on a Windows desktop. Source imports and a standard-library-only runtime support the dependency assessment but do not replace that exact environment check. Human group review, recordings and declaration remain pending. See Test Coverage for model-only measurement and its limits.